Import Libraries

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

Load Dataset

```
df = pd.read_csv("/content/sales_dataset csv.csv")
```

```
df.head()
```

	Order Date	Region	Sales	Year	Month	Month-Year	Profit
0	08-11-2017	South	261.9600	2017	11	Nov-17	52.3920
1	08-11-2017	South	731.9400	2017	11	Nov-17	146.3880
2	12-06-2017	West	14.6200	2017	6	Jun-17	2.9240
3	11-10-2016	South	957.5775	2016	10	Oct-16	191.5155
4	11-10-2016	South	22.3680	2016	10	Oct-16	4.4736

Next steps: Generate code with df New interactive sheet

Explore Dataset

```
print("\nDataset Info:")
df.info()
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Order Date  9800 non-null   object
 1   Region      9800 non-null   object
 2   Sales       9800 non-null   float64
 3   Year        9800 non-null   int64
 4   Month       9800 non-null   int64
 5   Month-Year  9800 non-null   object
 6   Profit      9800 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 536.1+ KB
```

Checking Missing Values

```
print("\nMissing values in each column:")
print(df.isnull().sum())
```

```
Missing values in each column:
Order Date    0
Region        0
Sales         0
Year          0
Month         0
Month-Year    0
Profit        0
dtype: int64
```

Statistical Summary of Dataset

```
print("\nBasic statistics:")
display(df.describe())
```

```
Basic statistics:
```

	Sales	Year	Month	Profit
count	9800.000000	9800.000000	9800.000000	9800.000000
mean	230.769059	2016.724184	7.818469	46.153812
std	626.651875	1.123984	3.281905	125.330375
min	0.444000	2015.000000	1.000000	0.088800
25%	17.248000	2016.000000	5.000000	3.449600
50%	54.490000	2017.000000	9.000000	10.898000
75%	210.605000	2018.000000	11.000000	42.121000
max	22638.480000	2018.000000	12.000000	4527.696000

Clean Dataset

```
df.fillna(0, inplace=True)          # Fill missing numeric values

df.drop_duplicates(inplace=True)    # Remove duplicate rows
```

Profit Calculation (If Missing)

```
if 'Profit' not in df.columns:
    if 'Sales' in df.columns and 'Cost' in df.columns:
        df['Profit'] = df['Sales'] - df['Cost']
    else:
        print("Profit cannot be calculated: Sales or Cost column missing")
```

Calculate Key Metrics

```
# Clean column names
df.columns = df.columns.str.strip()

# Convert to numeric
df['Sales'] = pd.to_numeric(df['Sales'], errors='coerce')
df['Profit'] = pd.to_numeric(df['Profit'], errors='coerce')

# Metrics
total_sales = df['Sales'].sum()
average_sales = df['Sales'].mean()
min_sales = df['Sales'].min()
max_sales = df['Sales'].max()
std_sales = df['Sales'].std()

total_profit = df['Profit'].sum()

if pd.notnull(total_sales) and total_sales != 0:
    profit_margin = (total_profit / total_sales) * 100
else:
    profit_margin = 0
print("Total Sales:", total_sales)
print("Average Sales:", average_sales)
print("Min Sales:", min_sales)
print("Max Sales:", max_sales)
print("Standard Deviation:", std_sales)
print("Total Profit:", total_profit)
print("Profit Margin:", profit_margin)
```

```
Total Sales: 2261152.7827000003
Average Sales: 230.87122551562183
Min Sales: 0.444
Max Sales: 22638.48
Standard Deviation: 626.8255665742824
Total Profit: 452230.55654
Profit Margin: 20.0
```

Statistical Measures

```
# Statistical Measures for Sales

sales_mean = df['Sales'].mean()
sales_median = df['Sales'].median()
sales_mode = df['Sales'].mode()[0]
sales_variance = df['Sales'].var()

print("Sales Statistics:")
print("Mean:", sales_mean)
print("Median:", sales_median)
print("Mode:", sales_mode)
print("Variance:", sales_variance)
```

```
Sales Statistics:
Mean: 230.87122551562183
Median: 54.593999999999994
Mode: 12.96
Variance: 392910.29091117013
```

Monthly Analysis

```
if 'Month' in df.columns:
    monthly_sales = df.groupby('Month')['Sales'].sum()
    monthly_profit = df.groupby('Month')['Profit'].sum()
    monthly_growth = monthly_sales.pct_change() * 100

    print("\nMonthly Sales:")
    display(monthly_sales)
    print("\nMonthly Profit:")
    display(monthly_profit)
    print("\nMonthly Growth (%):")
    display(monthly_growth)
```

```
Monthly Sales:
```

Month	Sales
1	94291.6296
2	59371.1154
3	197573.5872
4	135978.6686
5	154086.7237
6	145837.5233
7	145505.3610
8	157315.9270
9	300054.0717
10	199496.2947
11	350161.7110
12	321480.1695

dtype: float64

```
Monthly Profit:
```

Month	Profit
1	18858.32592
2	11874.22308
3	39514.71744
4	27195.73372
5	30817.34474
6	29167.50466
7	29101.07220
8	31463.18540
9	60010.81434
10	39899.25894
11	70032.34220
12	64296.03390

dtype: float64

```
Monthly Growth (%):
```

Month	Sales
1	NaN
2	-37.034586
3	232.777287
4	-31.175685
5	13.316835
6	-5.353609
7	-0.227762
8	8.116928
9	90.733435
10	-33.513219
11	75.522915
12	-8.190942

dtype: float64

Regional / Product Analysis

```
if 'Region' in df.columns:
    region_profit = df.groupby('Region')['Profit'].sum()
    print("\nProfit by Region:")
    display(region_profit)
```

```
Profit by Region:
```

Region	Profit
Central	98529.38264
East	133847.47080
South	77824.71420
West	142028.98890

dtype: float64

Loss Analysis

```
if 'Region' in df.columns:
    loss_data = df[df['Profit'] < 0]   # filter only loss rows
    region_loss = loss_data.groupby('Region')['Profit'].sum()

    print("\nLoss by Region:")
    display(region_loss)
```

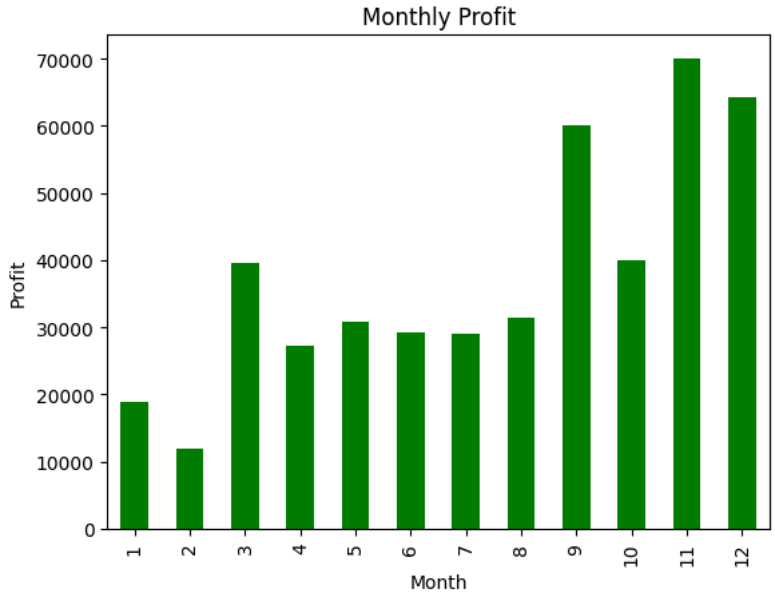
```
Loss by Region:
```

Region	Profit

dtype: float64

Monthly Visualization

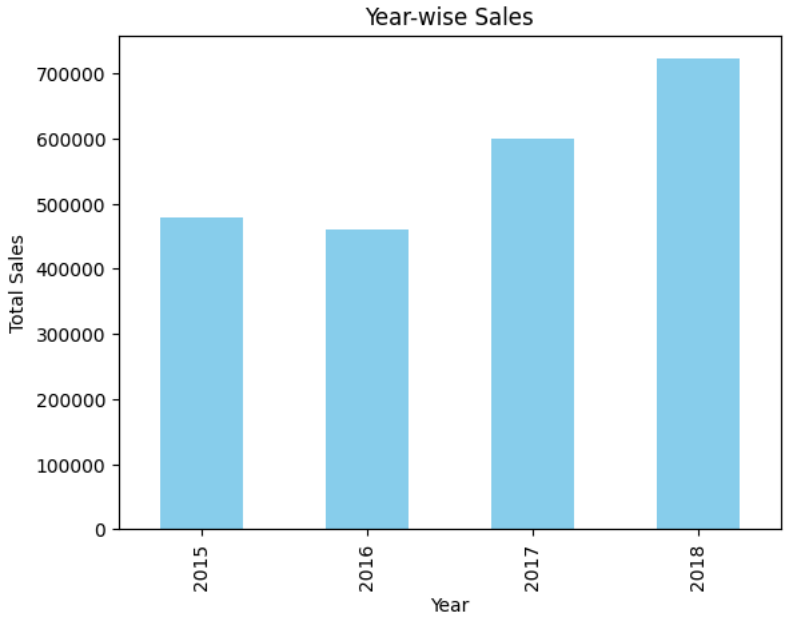
```
if 'Month' in df.columns:
    monthly_sales.plot(kind='line', marker='o', color='blue', title='Monthly Sales Trend')
    plt.xlabel('Month')
    plt.ylabel('Sales')
    plt.show()

    monthly_profit.plot(kind='bar', color='green', title='Monthly Profit')
    plt.xlabel('Month')
    plt.ylabel('Profit')
    plt.show()
```

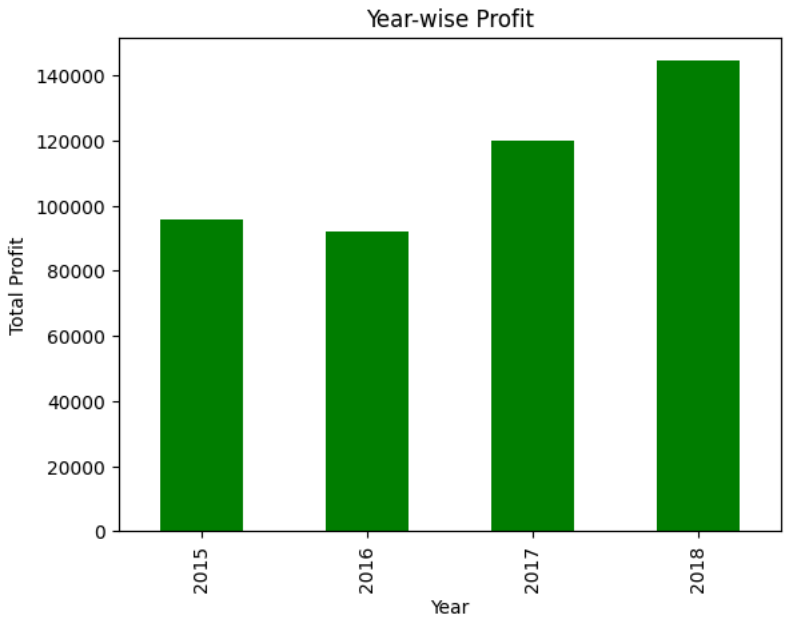


Year Wise analysis

```
# Year-wise Sales and Profit Analysis

yearly_sales = df.groupby('Year')['Sales'].sum()
yearly_profit = df.groupby('Year')['Profit'].sum()

print("Year-wise Sales:")
print(yearly_sales)

print("\nYear-wise Profit:")
print(yearly_profit)

# Plot yearly sales
yearly_sales.plot(kind='bar', color='skyblue')
plt.title("Year-wise Sales")
plt.xlabel("Year")
plt.ylabel("Total Sales")
plt.show()

# Plot yearly profit
yearly_profit.plot(kind='bar', color='green')
plt.title("Year-wise Profit")
plt.xlabel("Year")
plt.ylabel("Total Profit")
plt.show()
```

```
Year-wise Sales:
Year
2015    479551.8761
2016    459431.0774
2017    600137.2500
2018    722032.5792
Name: Sales, dtype: float64

Year-wise Profit:
Year
2015     95910.37522
2016     91886.21548
2017    120027.45000
2018    144406.51584
Name: Profit, dtype: float64
```

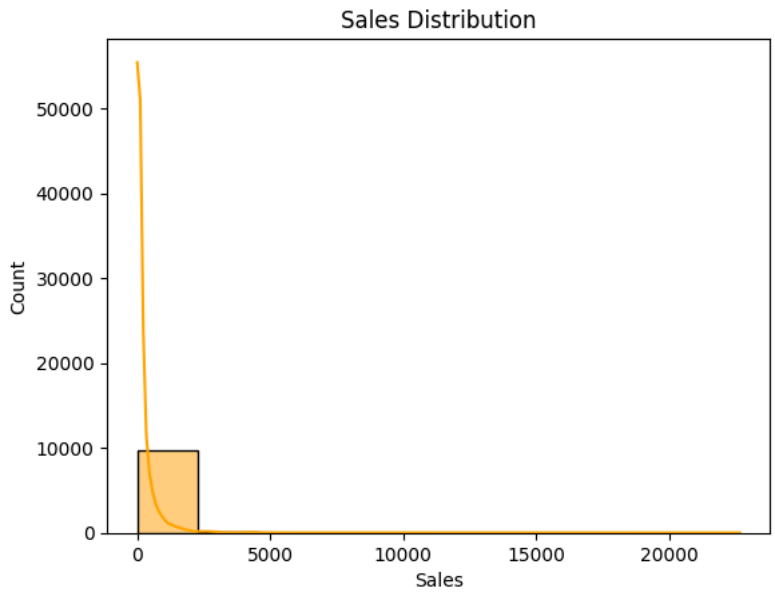


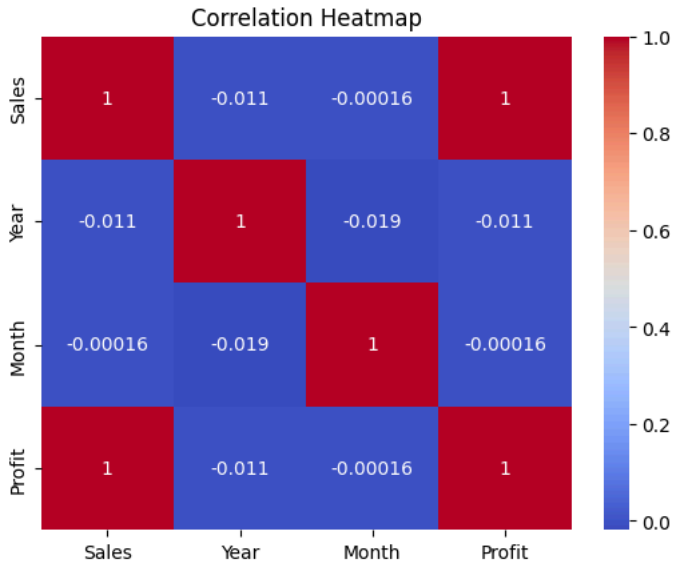


Data Visualization

```
# Sales Distribution
sns.histplot(df['Sales'], bins=10, kde=True, color='orange')
plt.title('Sales Distribution')
plt.show()

# Profit vs Sales Scatter Plot
sns.scatterplot(x='Sales', y='Profit', data=df)
plt.title('Profit vs Sales')
plt.show()

# Heatmap of numeric correlations
numeric_cols = df.select_dtypes(include=np.number)
sns.heatmap(numeric_cols.corr(), annot=True, cmap='coolwarm')
```

```
plt.title('Correlation Heatmap')
plt.show()
```





12. Save Cleaned & Processed Dataset

```
df.to_csv("/content/processed_sales_data.csv", index=False)
from google.colab import files
files.download("/content/processed_sales_data.csv")
```

Dashboard of Analysis

```
import pandas as pd
import matplotlib.pyplot as plt
import ipywidgets as widgets
from IPython.display import display, clear_output

# Load dataset (fixed file name)
data = pd.read_csv("/content/sales_dataset csv.csv")
data.columns = data.columns.str.strip()
# Convert to datetime
data["Order Date"] = pd.to_datetime(data["Order Date"], dayfirst=True)

# Create Month column (fixed issue)
data["Month"] = data["Order Date"].dt.strftime("%B")

# Create Profit/Loss status
data["Status"] = data["Profit"].apply(lambda x: "Profit" if x >= 0 else "Loss")

# Output widget
output = widgets.Output()

# ----------- FUNCTIONS -----------

def sales_summary(b):
    with output:
        clear_output()
        print(" SALES SUMMARY")
        print("Total Sales:", data["Sales"].sum())
        print("Average Sales:", data["Sales"].mean())
        print("Max Sales:", data["Sales"].max())
        print("Min Sales:", data["Sales"].min())

def profit_summary(b):
    with output:
        clear_output()
        print("PROFIT SUMMARY")
        print("Total Profit:", data["Profit"].sum())
        print("Average Profit:", data["Profit"].mean())
        print("Max Profit:", data["Profit"].max())
        print("Min Profit:", data["Profit"].min())

def profit_by_region(b):
    with output:
        clear_output()
        print(" PROFIT BY REGION")
        print(data.groupby("Region")["Profit"].sum())

def sales_by_region(b):
    with output:
        clear_output()
        print(" SALES BY REGION")
        print(data.groupby("Region")["Sales"].sum())

def profit_loss_status(b):
    with output:
        clear_output()
        print(" PROFIT vs LOSS")
        print(data["Status"].value_counts())

def monthly_sales(b):
    with output:
        clear_output()
        print("MONTHLY SALES TREND")
        monthly = data.groupby("Month")["Sales"].sum()
        print(monthly)

def show_table(b):
    with output:
        clear_output()
        print(" SAMPLE DATA")
        display(data.head(10))

# ----------- CHARTS -----------
```

```
def sales_chart(b):
    with output:
        clear_output()
        plt.figure()
        data.groupby("Region")["Sales"].sum().plot(kind='bar')
        plt.title("Sales by Region")
        plt.xlabel("Region")
        plt.ylabel("Sales")
        plt.xticks(rotation=45)
        plt.show()

def profit_chart(b):
    with output:
        clear_output()
        plt.figure()
        data.groupby("Region")["Profit"].sum().plot(kind='bar')
        plt.title("Profit by Region")
        plt.xlabel("Region")
        plt.ylabel("Profit")
        plt.xticks(rotation=45)
        plt.show()

def trend_chart(b):
    with output:
        clear_output()
        plt.figure()
```

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